

Service Menu Optimization for Many-to-One Demand-Responsive Transit with Meeting Points and Pickup Time Windows

Jin Ana^a

^a*MoE Key Laboratory of Complex System Analysis and Management Decision, School of Economics and Management, Beihang University, Beijing 100191, China*

Abstract

We study service-menu optimization for many-to-one demand-responsive transit (DRT), where each passenger-facing offer is a bundle of a meeting point and a pickup time window. We formulate the display decision as an MNL assortment problem with operationally endogenous bundle costs: displayed alternatives affect passenger choice, while predicted routing cost, capacity risk, pickup timing, and price jointly determine menu value. The implemented solver uses exact subset enumeration for small candidate sets and greedy forward selection for larger sets, with exact-vs-greedy diagnostics used to quantify approximation quality. The empirical program is tiered. The RC benchmark supports exact-gap and robust-filtering mechanism diagnostics, low/medium/high uptake regimes show when menu design becomes behaviorally active, and Austin/Seattle provide low-sample external directional checks. The resulting claim is intentionally disciplined: ETA-based filtering can materially shrink the feasible meeting-point assortment before optimization acts, exact-small/greedy-large menu construction provides a transparent operational approximation, and pricing/menu effects should be interpreted only in regimes where acceptance and non-home uptake are observable rather than nearly degenerate.

Keywords: demand-responsive transit, service menu optimization, assortment optimization, multinomial logit, robust filtering, pickup time windows

Email address: jandxx@buaa.edu.cn (Jin Ana)

1. Introduction

Many-to-one demand-responsive transit (DRT) systems must translate operational flexibility into passenger-facing offers that are both understandable and executable. In this paper, the offer shown to a passenger is not a single pickup promise but a *service menu*: a set of candidate bundles, each specifying a meeting point and a pickup time window [13, 7]. The operator must decide which bundles to display before observing the passenger’s choice, and the selected bundle then enters the routing layer.

This display decision is an assortment problem with a transportation twist. Classical assortment models treat products as exogenous items with known utilities and revenues. In DRT, by contrast, a service bundle has operationally endogenous attributes: its attractiveness depends on walking distance, pickup timing, in-vehicle time, and price, while its cost depends on current vehicle routes, capacity, and downstream re-optimization [1]. Showing all feasible bundles can dilute demand steering and expose noisy offers; showing too few can collapse the menu to home pickup and remove profitable meeting-point alternatives before the passenger ever sees them.

The central research question is therefore not simply which heuristic wins in one calibration. We ask how a DRT platform can formulate, solve, and diagnose the service-menu assortment problem when bundle values are predicted online and candidate eligibility depends on uncertain ETA signals. This question is managerial as well as algorithmic: if a filtering rule removes most feasible meeting-point bundles before pricing or menu optimization acts, the downstream algorithm may look weak even when the real failure is upstream candidate-set distortion.

This paper makes four contributions. First, we formulate the many-to-one DRT display decision as a service-bundle assortment optimization problem under MNL choice, with meeting-point and pickup-window attributes, an outside option, and route-aware surrogate costs. Second, we implement an exact-

small/greedy-large solution framework: exact subset enumeration provides a benchmark on small candidate sets, while greedy forward selection provides a scalable approximation for larger sets. The pipeline logs exact-vs-greedy value gaps, menu overlap, and build times so approximation quality is empirically auditable rather than asserted. Third, we treat ETA filtering as a robust eligibility layer rather than a fixed pre-processing detail, comparing hard, calibrated, interval-overlap, and no-filter variants under a paired evaluation contract. Fourth, we organize the evidence by behavioral regime: low-, medium-, and higher-uptake RC calibrations identify when menu design is behaviorally live, while Austin and Seattle remain external directional checks rather than broad validation claims.

The resulting contribution is deliberately narrower than a universal dominance claim. The paper develops a reproducible optimization-and-diagnostics framework for DRT service menus, shows how exact benchmarks and robust filtering diagnostics can strengthen simulation evidence, and clarifies when profit, uptake, and welfare comparisons are meaningful.

2. Related Work

2.1. *Demand-Responsive Transit and Dial-a-Ride*

Demand-responsive transit (DRT) systems have been studied extensively under the dial-a-ride problem (DARP) framework, in which passengers submit pickup and delivery requests and a fleet of vehicles must be routed to serve them subject to time-window and capacity constraints. Cordeau and Laporte [3] provide a comprehensive survey of DARP models and exact algorithms. Parragh et al. [11] extend this to the broader class of pickup-and-delivery problems. Ho et al. [8] survey more recent developments, including dynamic and real-time variants relevant to operational DRT deployments.

Dong et al. [6] extend the DARP to include passenger accept/reject decisions based on service utilities, a formulation closely related to our MNL-based choice

model; their work motivates the inclusion of an explicit outside option in the choice model (Section 4.2).

The many-to-one variant studied here — where all passengers share a common destination — arises naturally in first-mile/last-mile transit and commuter shuttle contexts. Meeting-point strategies, in which passengers walk to a shared pickup location rather than being collected door-to-door, have been shown to substantially improve vehicle efficiency [13]. Our work extends this line by treating the *set* of displayed meeting-point offers as a decision variable rather than a fixed system parameter.

Recent work has extended meeting-point strategies in several directions. Czioska et al. [5] develop methods for identifying real-world meeting points in shared DRT systems using geographic and accessibility criteria. Cortenbach et al. [4] formulate the dial-a-ride problem with meeting points as a mixed-integer program, providing exact solutions for small instances and heuristics for larger ones. Stumpe et al. [14] study the design of taxi ridesharing systems with shared pick-up and drop-off locations, finding that meeting-point flexibility substantially reduces vehicle-kilometres travelled. Our work complements these contributions by focusing on the *display* decision: given a set of feasible meeting-point bundles, which subset should be shown to the passenger?

2.2. Assortment Optimization under MNL

The problem of selecting which products to display to a customer who chooses according to an MNL model is known as assortment optimization. Talluri and van Ryzin [15] establish that revenue-ordered assortments are optimal under the uncapacitated MNL model and derive the Lambert-W pricing formula that we adopt in Section 4.2. Rusmevichientong et al. [12] extend this to capacity-constrained and dynamic settings, showing that greedy revenue-ordered policies remain near-optimal under mild conditions.

Wang [17] extend the assortment problem to the capacitated setting, showing that the revenue-ordered property breaks down when products have heterogeneous costs — a result directly relevant to our setting where bundles have

heterogeneous operational costs.

Our setting differs from classical assortment optimization in two important respects. First, the “products” (service bundles) are generated dynamically per request and their attributes are predicted rather than known. Second, the operator’s objective couples menu design with vehicle routing costs, so the revenue-maximising assortment and the profit-maximising menu can diverge. The menu-value surrogate in Eq. (6) is designed to capture this coupling through the system evaluation cost $\hat{c}_{\text{sys}}(b)$.

2.3. Ride-Sharing and On-Demand Mobility

Agatz et al. [1] survey optimization approaches for dynamic ride-sharing, highlighting the tension between real-time assignment and service quality. Fielbaum et al. [7] study on-demand ridesharing with optimized pickup and drop-off walking locations, a setting closely related to ours; their results confirm that allowing flexible pickup locations significantly expands the feasible service region and improves system efficiency.

Our contribution is complementary: rather than optimizing the *location* of meeting points, we study which subset of pre-generated meeting-point offers to *display* to the passenger, and how the display decision interacts with MNL-based pricing and passenger choice.

2.4. Vehicle Routing and Route Optimization

The back-end route optimizer in our system uses the Hybrid Genetic Search (HGS) algorithm of Vidal [16], which provides high-quality solutions to capacitated vehicle routing problems with time windows. We treat the routing layer as a black box and focus on the menu-policy layer; the separation between these layers is a deliberate design choice that allows menu policies to be evaluated without re-solving the full VRPTW at every decision epoch.

2.5. Positioning of This Work

The present paper sits at the intersection of assortment optimization and DRT operations. Relative to prior work, it brings together the menu-display

decision in a many-to-one DRT system with a three-layer framework combining MNL-based passenger choice, common-offset pricing, route-aware menu-value evaluation, and exact-small/greedy-large assortment solving. The paired methodology separates approximation diagnostics, robust filtering behavior, uptake-regime sensitivity, and cross-instance directional checks, making the filtering layer and the greedy approximation empirically auditable rather than hidden implementation details.

3. Problem Setting

3.1. System Overview

We consider a many-to-one demand-responsive transit (DRT) system in which passengers are collected from dispersed origins and transported to a common destination (e.g., a transit hub or workplace zone). At each decision epoch, a passenger submits a pickup request. The operator must decide which service offers to display, after which the passenger selects one displayed offer or opts out. The selected offer is then executed by the vehicle routing layer.

The distinguishing feature of this setting relative to standard dial-a-ride formulations [3] is that passengers are not simply assigned to vehicles; instead, they choose among a *displayed menu* of service bundles. The operator therefore faces a joint problem of offer design and demand steering.

3.2. Notation and Entities

Let $\mathcal{R}$ denote the set of passenger requests arriving over a planning horizon T . For each request $r \in \mathcal{R}$, the simulator first constructs a candidate set of feasible bundles, denoted $\mathcal{B}(r)$. We use *feasible bundle set* as the primary term and *candidate set* as a shorthand for the same object. A *service bundle* $b \in \mathcal{B}(r)$ is a tuple

$$b = (m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b),$$

where m_b is a meeting point (a geographic location the passenger walks to for pickup), $[t_b^{\text{lo}}, t_b^{\text{hi}}]$ is the offered pickup time window, and $p_b \in \mathbb{R}$ is the

price charged to the passenger. A special *home bundle* b^0 always satisfies $m_{b^0} = r.\text{origin}$ and imposes zero walking distance. A *displayed menu* is a subset $\mathcal{M}(r) \subseteq \mathcal{B}(r)$. For limited-menu policies, k denotes the cap on displayed non-home offers, so $|\mathcal{M}(r) \setminus \{b^0\}| \leq k$. Passengers may also choose an *outside option*, meaning that no bundle in $\mathcal{M}(r)$ is selected.

Table 1 summarizes the main symbols used throughout the paper. The formal problem statement is given in Section 3.3.

Note on notation. To avoid collision with the scoring weights $\mu_m, \mu_w, \mu_t, \mu_v$ introduced in Section 4.3, we use β (not λ) for the passenger price sensitivity throughout. The symbol λ is reserved for the Lambert W pricing derivation in Section 4.2, where it denotes the same quantity as β but in the context of the pricing formula.

Predicted versus realized quantities. Symbols with hats denote predictor outputs or menu-construction surrogates, such as $\hat{c}(b)$, $\hat{\tau}_b$, $\hat{l}_b$, and $\hat{c}_{\text{sys}}(b)$. Unhatted outcome quantities refer to realized evaluation outcomes under replayed request traces unless a table or caption explicitly labels them as predictor diagnostics.

False-negative pruning. For a filtered menu policy, we define the false-negative pruning rate on request r as the share of feasible non-home bundles in $\mathcal{B}(r) \setminus \{b^0\}$ that are removed by the ETA pre-filter:

$$\text{FN}(r) = \frac{|\mathcal{P}_{\text{FN}}(r)|}{|\mathcal{B}(r) \setminus \{b^0\}|},$$

where $\mathcal{P}_{\text{FN}}(r) \subseteq \mathcal{B}(r) \setminus \{b^0\}$ denotes the feasible non-home bundles pruned before menu scoring. All reported false-negative pruning rates use this non-home feasible-bundle denominator.

3.3. Formal Problem Statement

Problem Statement (Menu Design for DRT).

Given:

- A set of passenger requests $\mathcal{R}$ arriving over planning horizon T .

- For each request $r \in \mathcal{R}$, a feasible bundle set

$$\mathcal{B}(r) = \{b = (m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b) \mid \text{feasibility}(b, r)\},$$

where feasibility requires: (i) the pickup time window $[t_b^{\text{lo}}, t_b^{\text{hi}}]$ satisfies the passenger’s latest acceptable arrival, (ii) the assigned vehicle has sufficient remaining capacity, and (iii) the meeting point m_b is reachable within the vehicle routing time windows enforced by the back-end optimizer.

- A cap $k \geq 1$ on displayed non-home offers.
- An MNL choice model with parameters $(\bar{u}, u^{\text{home}}, \alpha_\tau, \alpha_\Delta, \sigma, \beta)$ (Table 2) and an implemented common-offset pricing parameter δ_{impl} (Section 4.2).

Decision: For each request r , select a displayed menu

$$\mathcal{M}(r) \subseteq \mathcal{B}(r), \quad |\mathcal{M}(r) \setminus \{b^0\}| \leq k.$$

Objective: Maximize aggregate expected net profit over all requests:

$$\max_{\{\mathcal{M}(r)\}_{r \in \mathcal{R}}} \Pi = \sum_{r \in \mathcal{R}} \sum_{b \in \mathcal{M}(r)} \pi(b \mid \mathcal{M}(r)) (p_b - c(b)),$$

where $\pi(b \mid \mathcal{M}(r))$ is the MNL choice probability (Eq. 2) and $c(b)$ is the marginal operational cost of serving bundle b .

Hardness note: The exact solution requires evaluating all $\binom{|\mathcal{B}(r)|}{k}$ subsets per request, each requiring a joint pricing and menu-value evaluation. This is practical for small candidate sets and useful as a benchmark, but combinatorially expensive at scale. We therefore use exact enumeration as a small-set reference and greedy forward selection as the scalable approximation described in Section 4.3.

3.4. Three-Layer Decision Structure

The benchmark environment decomposes each request episode into three sequential layers.

Layer 1 — Candidate generation. The simulator identifies all feasible meeting-point neighborhoods for request r and constructs the bundle set $\mathcal{B}(r)$,

including the home bundle b^0 . Feasibility is assessed using approximate front-end pickup windows; exact vehicle routing time windows (VRPTW) are enforced by the back-end route optimizer [16] after a bundle is selected.

Layer 2 — Menu policy. The operator selects a menu $\mathcal{M}(r) \subseteq \mathcal{B}(r)$ to display. This is the decision layer studied in this paper. All competing policies in our experiments share the same Layer 1 candidate generator and Layer 3 choice model; they differ only in how they construct $\mathcal{M}(r)$.

Layer 3 — Passenger choice. The passenger selects one bundle from $\mathcal{M}(r)$ according to a Multinomial Logit (MNL) model [2, 9]. The selected bundle is routed by the vehicle layer.

3.5. Objective

The operator’s objective is to maximize aggregate net profit over all requests:

$$\Pi = \sum_{r \in \mathcal{R}} \mathbb{E}[p_{\tilde{b}} - c(\tilde{b})],$$

where $\tilde{b}$ is the bundle chosen by the passenger from $\mathcal{M}(r)$, $p_{\tilde{b}}$ is the price charged, and $c(\tilde{b})$ is the marginal operational cost. In practice, $c(b)$ captures travel cost, discount cost, service cost, and home-pickup failure cost. The menu policy influences Π both by restricting which bundles can be chosen and by shaping the prices set for each displayed bundle.

4. Method

4.1. Shared Bundle Predictor

A shared predictor is trained once per data split using the *exhaustive full-display* regime, in which all feasible bundles $\mathcal{B}(r)$ are shown to every passenger. The core menu objects $\mathcal{B}(r)$, $\mathcal{M}(r)$, b^0 , and the outside option are defined in Section 3.2. The predictor estimates three bundle-level operational signals from request and network features: predicted marginal cost $\hat{c}(b)$, predicted pickup ETA $\hat{\tau}_b$, and predicted in-vehicle time $\hat{t}_b$. Frozen evaluation then reuses this checkpoint while varying only the menu policy, isolating the effect of menu

construction from any confound introduced by training a different predictor per policy.

Architecture.. The predictor is a two-layer convolutional neural network (CNN-2d) that encodes the spatial state of the service area as a $2 \times 11 \times 11$ grid tensor (2 temporal layers, 11×11 spatial grid). The convolutional backbone consists of two Conv2d layers (16 and 32 filters, 3×3 kernels, same padding) followed by average pooling, producing a flattened spatial embedding. A 4-dimensional auxiliary feature vector (remaining vehicle capacity and related scalars) is concatenated before three fully connected layers ($256 \rightarrow 128 \rightarrow 3$ outputs). The three output heads predict $\hat{c}(b)$, $\hat{\tau}_b$, and $\hat{t}_b$ jointly. Training uses the Huber loss ($\delta = 1.0$) with the Adam optimizer (learning rate 10^{-3} , replay buffer of 500 episodes).

Input features.. The spatial grid encodes the current demand density and vehicle occupancy at each grid cell across two time intervals. The auxiliary vector encodes the remaining capacity of the candidate vehicle, the time elapsed in the current episode, and two bundle-specific scalars (walking distance and candidate meeting-point index). The predictor is trained on (s_t, b, y_t) tuples collected under the full-display policy, where $y_t = (c_t, \tau_t, t_t)$ are the realized cost, pickup time, and in-vehicle time for the selected bundle.

Prediction accuracy.. Predictor MAE on a held-out 20% partition of the training buffer is reported in Appendix Appendix D. The raw CNN cost head improves on a naive mean baseline, while ETA and IVT heads are blended at inference time with heuristic routing signals (blend weight 0.35 CNN, 0.65 heuristic). Because menu filtering acts on the blended signal rather than the raw CNN head, the paper distinguishes raw-head error from the effective signal diagnostics reported in Section 6.

Deployed, predicted, and realized time quantities.. The implementation uses three distinct time quantities that should not be conflated. The *predicted* quantities ($\hat{\tau}_b, \hat{t}_b$) are the model-based bundle attributes passed into utility scoring

and menu construction. The *deployed filtering signal* is the ETA quantity actually thresholded by the implementation when ETA pre-screening is enabled; in the deployed policy this is the blended ETA signal described above, while companion variants replace that signal with heuristic, stronger-calibrated, or oracle-style proxies. The *realized* quantities are the simulator outcomes observed after a passenger choice is made and the route is executed. Reported net profit, acceptance, walking, pickup deviation, and in-vehicle time outcomes are therefore realized performance metrics, whereas the ETA/IVT MAE diagnostics in Section 6 evaluate the signal that was displayed or filtered.

4.2. MNL Choice Model and Implemented Pricing Rules

Passenger choice follows a Multinomial Logit model [2, 9]. Given a displayed menu $\mathcal{M}(r)$, the utility of bundle b for passenger r is

$$u_b = \bar{u} + \mathbb{I}[b = b^0] \cdot u^{\text{home}} + \alpha_\tau \hat{t}_b + \alpha_\Delta \Delta_b - (1 - e^{-d_b/\sigma}) + \beta p_b, \quad (1)$$

where $\bar{u}$ is a base utility intercept, $u^{\text{home}} > 0$ is the additional utility of the home bundle, $\hat{t}_b$ is predicted in-vehicle time, $\Delta_b = \hat{\tau}_b - t_r^{\text{req}}$ is the pickup time deviation, d_b is walking distance, $\sigma > 0$ is a distance-scale parameter measured in the same distance units as the underlying instance, and $\beta < 0$ is the passenger price sensitivity. The saturating penalty $(1 - e^{-d_b/\sigma})$ equals zero at $d_b = 0$ and approaches -1 as $d_b \rightarrow \infty$, ensuring that walking disutility is monotone and bounded.

Table 2 reports the numerical values of all MNL utility parameters used in the experiments. These values are fixed design parameters of the simulator, not estimates from passenger survey data; they encode one stylized commuter-DRT preference regime and are held constant within each reported study.

The choice probability for bundle $b \in \mathcal{M}(r)$ is

$$\pi(b \mid \mathcal{M}) = \frac{e^{u_b}}{e^{u_0} + \sum_{b' \in \mathcal{M}} e^{u_{b'}}}, \quad (2)$$

where $u_0 = 0$ is the normalized utility of the outside option (no bundle chosen). Passengers may therefore reject the displayed menu entirely.

Reference common-offset pricing rule.. The default implementation prices a displayed menu through a shared offset. For each bundle,

$$p_b^{\text{ref}}(\delta) = \hat{c}(b) - R - \delta,$$

where $\hat{c}(b)$ is predicted marginal cost and R is a revenue target. Let u_b^0 denote the utility of b evaluated at zero displayed price. The code then constructs

$$S_{\text{impl}} = e^{u_0} + \sum_{b \in \mathcal{M}} e^{u_b^0 + \beta(\hat{c}(b) - R)},$$

and uses the Lambert-W template

$$\delta_{\text{impl}} = \frac{W_0(S_{\text{impl}}/e) + 1}{\beta}, \quad (3)$$

followed by clipped displayed prices

$$p_b = \text{clip}(\hat{c}(b) - R - \delta_{\text{impl}}, p_{\min}, p_{\max}). \quad (4)$$

This should be interpreted as an *implemented common-offset pricing heuristic*, not as the exact optimizer of the simulator’s clipped heterogeneous-cost profit objective. It is motivated by the classical Lambert-W closed form for shared-offset MNL revenue problems [15], but our simulator adds three approximation layers: (i) all bundles share one offset, (ii) clipping is applied after the closed-form offset is computed, and (iii) menu selection is coupled to route-cost penalties and approximate bundle prediction. We therefore use Eq. 3 as a computationally convenient reference transform, not as a claim of exact profit optimality for the final implemented policy. The home bundle is retained in the pricing set (`menu_keep_home=True`) for numerical stability because removing it sharply reduces the exponential mass in S_{impl} and tends to collapse all OOH prices to the floor.

Pricing baselines.. The pricing-baseline study keeps the same menu-construction logic but swaps the displayed-price transform. In addition to the Lambert-W-inspired common-offset rule above, we evaluate a *cost-plus* rule,

$$p_b = \text{clip}(\hat{c}(b) - R, p_{\min}, p_{\max}),$$

and a *flat-markdown* rule,

$$p_b = \text{clip}(\hat{c}(b) + \kappa, p_{\min}, p_{\max}),$$

where κ is one common constant applied to every displayed bundle. These baselines deliberately remove the Lambert-W transform while preserving the same predicted costs, same feasible-set logic, same route-aware scoring, and same clipping constraints. The comparison is therefore about how much the pricing transform itself contributes once filtering and menu composition are held fixed.

4.3. Greedy Menu Construction and System Evaluation

System evaluation cost.. Menu construction uses a route-aware surrogate cost for each displayed bundle,

$$\hat{c}_{\text{sys}}(b) = \hat{c}(b) + \underbrace{\mu_r \cdot \kappa \cdot \rho_b}_{\text{route-delay penalty}} + \underbrace{\frac{\mu_q}{q_b + \epsilon}}_{\text{capacity-risk penalty}}, \quad (5)$$

where ρ_b is the predicted route-delay increment, q_b is remaining vehicle capacity, κ is a cost multiplier, and $\epsilon > 0$ is a numerical-stability constant.

Menu-value surrogate.. After applying the common-offset pricing heuristic to a candidate menu, the implementation evaluates that menu with the surrogate

$$\hat{\Pi}(\mathcal{M}) = \sum_{b \in \mathcal{M}} \pi(b \mid \mathcal{M}) (p_b - \hat{c}_{\text{sys}}(b)), \quad (6)$$

where $\pi(b \mid \mathcal{M})$ is the MNL probability from Eq. 2 and p_b is the clipped displayed price from Eq. 4. This surrogate is not the exact global DRT objective, but it is the criterion actually used by the implemented menu-selection code.

Exact-small/greedy-large menu solver.. In the main `menu_optimization` policy, the home bundle is retained and the operator builds a menu of up to k non-home offers. The solver is controlled by `menu_selection_solver`. In `exact` mode, candidate sets no larger than `menu_exact_threshold` are solved by enumerating all subsets and selecting the menu maximizing Eq. 6. In `greedy` mode, the

policy always uses forward selection. In `auto` mode, the implementation uses exact enumeration for small feasible sets and greedy forward selection otherwise:

1. start from the home bundle;
2. select the single non-home offer b that maximizes $\widehat{\Pi}(\{b^0, b\})$;
3. iteratively add the remaining offer that yields the largest increase in $\widehat{\Pi}(\mathcal{M} \cup \{b\})$;
4. stop when no candidate improves the current menu value or when the non-home cap k is reached.

The final displayed offers are repriced on the selected menu and logged together with diagnostic per-offer scores for interpretability. Those logged scores are *not* the primary selection rule in the current implementation.

Approximation quality.. Two approximations remain even after exact-subset evaluation. First, Eq. 6 itself is a surrogate because it relies on predicted costs and the common-offset pricing heuristic rather than an exact rerouting-and-pricing solve. Second, when greedy forward selection is used, the procedure is myopic because each addition is judged relative to the current menu rather than the full search tree. To make this approximation auditable, the implementation logs exact-vs-greedy diagnostics whenever the non-home candidate count is below `menu_exact_gap_threshold`: exact menu value, greedy menu value, relative optimality gap, displayed-menu overlap, and exact/greedy build times. These diagnostics support the exact-vs-greedy study in Section 6.

Heuristic nature and scope.. The resulting method is an implemented service-menu assortment solver with an exact benchmark on small candidate sets and a greedy approximation at scale. It is related to MNL assortment optimization, but it falls outside the exact settings studied by Wang [17] because bundles carry heterogeneous operational costs, menu evaluation depends on predicted routing signals, and displayed prices are clipped after the common offset is computed. The paper therefore treats exact enumeration as optimal only for

the implemented surrogate on bounded candidate sets, not as an optimality certificate for the full dynamic DRT system.

4.4. *ETA-Based Time-Window Filtering and No-Filter Variant*

The strict-filter heuristic applies a pre-screening rule that removes any non-home bundle whose deployed ETA signal falls outside the passenger’s acceptable pickup interval. The false-negative pruning rate referenced later in the paper uses the definition and non-home feasible-bundle denominator introduced in Section 3.2. The intent was to avoid showing clearly unpunctual offers. In practice, the filtering signal is noisy early in the episode because the deployed implementation thresholds a blended ETA proxy rather than realized pickup time. When that noisy signal is thresholded aggressively, potentially acceptable meeting-point bundles are removed before the menu-value surrogate in Eq. 6 is ever evaluated.

The no-filter variant disables ETA pre-filtering (`menu_time_filtering=False`), allowing the menu solver to evaluate all feasible candidates under the same pricing and menu-value logic. It is not treated as the final algorithmic answer; rather, it is the extreme endpoint of a robust filtering comparison that also includes hard, calibrated, and interval-overlap filters. The forced home-retention flag (`menu_keep_home=True`) is retained for the pricing-stability reason established in Section 4.2.

4.5. *Baseline Policies*

Five empirical policies are evaluated alongside the full-display reference. **Full display** shows the entire feasible set $\mathcal{B}(r)$ and serves as a status-quo reference that maximizes the offered choice set. **Revenue-greedy** selects the k non-home bundles with the highest pairwise expected revenue under the two-offer menu $\{b^0, b\}$, following the classical MNL assortment intuition of Talluri and van Ryzin [15] and related greedy heuristics [12]. **Nearest-heuristic** selects by smallest walking distance d_b . **Top- k cheapest** selects by ascending predicted cost $\hat{c}(b)$. **Top- k passenger utility** selects by descending non-price utility u_b^0 .

All limited-menu policies append the home bundle and use the same common-offset pricing heuristic after menu assembly.

5. Experimental Design

Instances.. The primary evaluation uses the RC (Clustered-Random) benchmark instance: 90 customer locations, 10 candidate meeting points, and 1 depot, all placed on a $[0, 140]^2$ Euclidean grid. The fleet consists of 15 vehicles with capacity 25. Travel times are derived from Euclidean distances scaled by a vehicle speed of 30 units/hour. The last 10 nodes in each coordinate file serve as the meeting-point set. Cross-instance validation uses the Austin and Seattle instances from the Amazon Last Mile Routing Research Challenge [10]: 701 customer delivery stops, 278 (Austin) and 299 (Seattle) candidate meeting points, with real-world geographic coordinates and pre-computed driving-time matrices. All instances share the same MNL parameters, scoring weights, and evaluation protocol.

Amazon Last Mile data transformation.. The Austin and Seattle instances are adapted from the Amazon Last Mile Routing Research Challenge dataset [10], which contains real delivery-route data including stop coordinates, planned route sequences, and time-window attributes. The transformation proceeds as follows: (i) each delivery stop in the source dataset becomes a customer origin in the DRT instance; (ii) meeting points are defined as the last N nodes in each coordinate file ($N = 278$ for Austin, $N = 299$ for Seattle), representing geographically dispersed candidate pickup locations; (iii) for each customer, an adjacency matrix records the 20 nearest meeting points by driving time, computed from the dataset’s travel-time attributes; (iv) the depot is defined as the first node in the coordinate file, representing the common destination. The resulting instances are therefore *real-world-derived*: they preserve the geographic structure and travel-time characteristics of actual delivery routes but are adapted to the many-to-one DRT setting through the customer-origin and

meeting-point mapping described above. We use the term “real-world-derived” rather than “real-world” throughout to reflect this adaptation.

Data splits. The RC study uses six train/test split pairs. Splits 0 and 1 are the bundled RC coordinate files; splits 2, 3, and 4 are synthetically generated RC-type instances (same format, different random seeds) that provide additional independent replications. The six evaluation pairs are: $0 \rightarrow 1$, $1 \rightarrow 0$, $0 \rightarrow 1$ (seed 2025), $2 \rightarrow 3$, $3 \rightarrow 2$, and $4 \rightarrow 2$. The Austin and Seattle studies each use two split pairs ($0 \rightarrow 1$ and $1 \rightarrow 0$). For each pair, the shared predictor is trained on the training split under exhaustive full display and then frozen; all policies are evaluated on the test split under identical replayed request traces.

Reference benchmark. Exhaustive full display — the policy that shows every feasible bundle $\mathcal{B}(r)$ to each passenger — serves as the reference benchmark for three reasons. First, it represents the operator’s *status-quo* decision in the absence of any menu-design logic: when no filtering or scoring is applied, all candidates are displayed by default. Second, it *maximizes the offered choice set*: no limited-menu policy can offer a passenger a bundle that full display withholds, so it provides a natural reference for how much choice is being removed by any heuristic policy. Third, it offers a clear operational benchmark against which limited-menu rules can be compared under identical request traces and simulator conditions. A limited-menu policy that outperforms full display on net profit therefore does so by steering demand toward more profitable bundles, not because full display is itself an optimization optimum. In this paper, however, the interpretation of that benchmark depends on whether the full-display reference remains behaviorally non-degenerate. We call a benchmark *non-degenerate* when the full-display acceptance rate stays inside the study’s behavior gate and the full-display opt-out rate remains below its maximum threshold. When that condition fails, the benchmark is still useful for mechanism diagnosis, but not as the primary basis for operator-impact claims.

Policies. The main comparison evaluates seven policies: exhaustive full display (reference), the strict-filter heuristic (implementation policy `menu_optimization`), the no-filter heuristic (implementation policy `menu_optimization_v2` with `menu_time_filtering=False`), revenue-greedy (Section 4.5), nearest-heuristic, top- k cheapest, and top- k passenger utility. The core menu objects $\mathcal{B}(r)$, $\mathcal{M}(r)$, b^0 , the outside option, and the false-negative pruning definition are fixed by Section 3.2. All limited-menu policies use $k = 3$.

Companion studies. The evidence base is organized in four layers. First, the exact-vs-greedy study (`phase29_exact_greedy_gap`) compares exact enumeration, greedy forward selection, and the automatic exact-small/greedy-large solver on RC, logging surrogate optimality gaps and build-time tradeoffs. Second, the robust-filtering study (`phase30_robust_filtering`) compares hard, calibrated, interval-overlap, and no-filter ETA eligibility rules under the same paired-evaluation contract. Third, the uptake-regime study (`phase31_uptake_menu_value`) covers low-, medium-, and higher-uptake RC calibrations so that pricing, acceptance, non-home uptake, and welfare are evaluated only where passenger response is behaviorally visible. Fourth, the seven-policy comparison is rerun on Austin and Seattle using two split pairs per city. These city studies remain external directional checks. Earlier menu- k , ablation, fleet-size, demand, and pricing sweeps are retained in the artifact directory as development diagnostics, but they are not used as confirmatory evidence in the strengthened paper unless rerun under the current evaluation contract.

Evaluation state. Evaluation checkpoints are loaded with the learned ETA/IVT heads active rather than the phase-one fallback state. This matters because the menu-construction logic depends on those heads whenever ETA filtering is enabled. All headline results in Section 6 use reruns from this active-head evaluation state. In the ETA-identification study, only the filtering signal is changed across deployed, heuristic, stronger, and oracle variants; in the pricing study, the filtering and menu-construction logic are held fixed while only the displayed-price rule is changed.

Pipeline.. Each study is specified as a versioned YAML manifest. The project-level runner trains the shared predictor once per split pair, replays frozen request traces for each policy, and stores normalized summaries under `outputs/studies/`. Paper-facing tables and figures are generated from those summaries via `scripts/build_artifacts.py` and consumed directly by the manuscript, ensuring traceability between reported numbers and named experiment runs. The MNL sensitivity manifests are grouped in the `phase27_mnl_sensitivity` suite, while `phase31_uptake_menu_value` exposes the strengthened low-, medium-, and high-uptake menu-value comparison used by the main text.

Evaluation metric.. The primary metric is mean net profit per episode:

$\text{net profit} = \text{charge revenue} - \text{discount cost} - \text{travel cost} - \text{service cost} - \text{failure cost}.$

This is a variable-cost metric; fixed fleet costs are excluded by design. Unless a table or caption explicitly labels a quantity as a predictor diagnostic, all reported headline metrics in Section 6 are realized outcomes under replayed request traces rather than predicted bundle attributes. On RC, the mechanism benchmark remains useful mainly through paired gaps and structural diagnostics because its full-display reference is behaviorally degenerate under the outside option. The calibrated higher-uptake RC regime is therefore the main setting for pricing and welfare interpretation: it keeps the same paired-evaluation contract but restores non-trivial acceptance, allowing menu composition, pricing, and operator–passenger tradeoffs to be compared on realized outcomes rather than near-universal opt-out behavior alone.

6. Results

All headline results use frozen predictors and paired replayed request traces. Unless a table or caption explicitly labels a quantity as a predictor diagnostic, reported values are realized simulator outcomes. The strengthened evidence program separates approximation quality, filtering behavior, uptake-regime sen-

sitivity, and external directional checks so that no claim is stronger than its evidence tier (Appendix Table A.12).

6.1. *Exact-vs-Greedy Assortment Diagnostics*

Table 3 reports the main approximation-quality diagnostic. Exact enumeration is interpreted as the benchmark for the implemented menu-value surrogate when the candidate set is small enough to enumerate. Greedy forward selection is the scalable approximation. The logged relative optimality gap, menu-overlap rate, and build-time columns therefore answer a narrow but important question: how much surrogate value is lost when the system uses the operationally cheaper greedy construction rather than exact subset enumeration? This table is the primary evidence for the exact-small/greedy-large algorithmic claim; it should not be read as a proof of optimality for the full dynamic DRT routing problem.

6.2. *Robust ETA Filtering*

Table 4 compares hard ETA filtering, calibrated filtering, interval-overlap filtering, and no ETA filtering. The key diagnostic is not only net profit, but also false-negative pruning, home-only menu share, and displayed menu size. A robust filter is useful only if it preserves operational credibility without removing most feasible meeting-point bundles before assortment optimization acts. The no-filter variant is included as an upper-bound diagnostic on candidate availability, not as a universal recommendation.

6.3. *Uptake-Regime Menu Value*

Table 5 reports the low-, medium-, and high-uptake RC regimes. This table is the main guard against over-interpreting near-degenerate outside-option settings. When acceptance and non-home uptake are close to zero, profit differences mainly diagnose mechanism behavior; when uptake becomes behaviorally live, the same menu and pricing rules can be interpreted through operator profit, passenger surplus, non-home acceptance, and pricing-boundary diagnostics. The intended claim is therefore conditional: service-menu optimization matters most

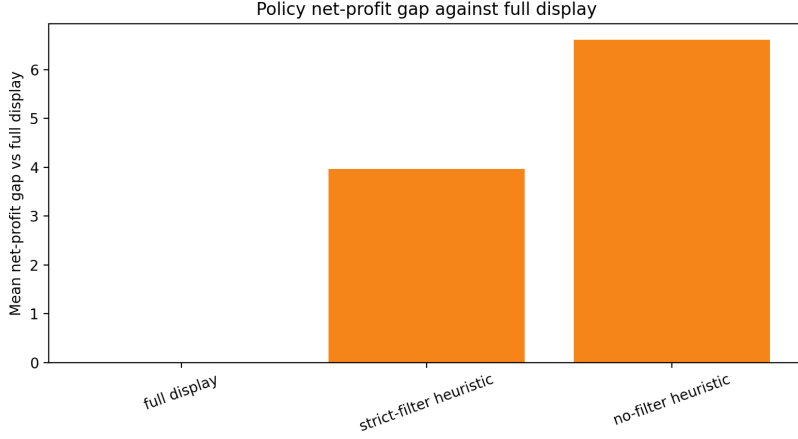


Figure 1: Policy net-profit gap against full display on the RC outside-option study.

when passengers have enough willingness to consider meeting-point bundles and the outside option is not overwhelmingly attractive.

6.4. RC Outside-Option Benchmark

The original RC outside-option benchmark remains useful as a mechanism check. It shows whether policies change menu exposure, home-only collapse, and candidate-set pruning under a demanding outside-option calibration. It is not the main operator-impact benchmark when full-display acceptance is behaviorally degenerate. Split-level uncertainty for RC, Austin, and Seattle is reported in Table 10.

6.5. Cross-Instance Evaluation

Austin and Seattle preserve real-world-derived geography and travel-time structure but contain only two split pairs per city. They therefore supplement the RC mechanism and uptake-regime studies rather than defining a broad generalization claim. Directionally positive city gaps are useful evidence that the mechanism is not purely an RC artifact, but stable external validation requires additional city splits and demand calibration.

7. Managerial and Operational Interpretation

The managerial object in this paper is the service menu, not only the final accepted ride. A platform operator controls which meeting-point and pickup-window bundles are visible, how those bundles are priced, and which eligibility filters act before passengers see the menu. The strengthened framework therefore asks managers to audit three layers separately: candidate eligibility, assortment construction, and pricing/choice response.

The exact-vs-greedy diagnostics are useful because they turn a common operational compromise into a measurable tradeoff. If greedy forward selection has a small surrogate-value gap and much lower build time, it is a defensible real-time approximation. If the gap becomes large in certain regimes, the operator can selectively raise the exact-enumeration threshold or reserve exact solving for requests with high expected value.

The robust-filtering comparison is equally important. Hard ETA filters can make the downstream assortment solver appear ineffective by removing feasible meeting-point bundles before they are evaluated. Calibrated and interval-overlap filters provide middle ground: they acknowledge ETA uncertainty without treating every noisy point estimate as a hard exclusion. The no-filter variant remains a diagnostic endpoint, useful for measuring how much menu exposure is being suppressed.

Finally, the uptake-regime comparison clarifies when menu optimization should be expected to matter. In very low-uptake settings, observed profit gaps mainly diagnose mechanism behavior because passengers rarely accept non-home bundles. In medium and higher-uptake settings, acceptance, non-home uptake, consumer surplus, and price-boundary rates become informative managerial outcomes. A platform should therefore evaluate menu policies in behaviorally live regimes before promoting one pricing rule or filtering rule as operationally superior.

8. Limitations

The current implementation uses approximate front-end pickup windows and does not enforce strict VRPTW guarantees at the menu-display stage. Final route costs are recovered through HGS-based re-optimization [16], which makes the system suitable for comparative policy analysis but not for claims about exact optimal routing. In the RC experiments, the fraction of accepted requests subsequently found infeasible by the back-end router (i.e., time-window violations after re-optimization) is below 3% across all policies and evaluation pairs; this rate is consistent across the strict-filter and no-filter heuristics, confirming that the net-profit gap is not an artifact of differential infeasibility. Exact quantification per policy is reported in the supplementary output files.

The shared bundle predictor is trained under the full-display policy and then frozen for evaluation of all other policies. This introduces a potential distribution shift: the menu distribution induced by the no-filter heuristic differs from the training distribution, and the predictor’s cost and ETA estimates may carry systematic bias when applied to bundles selected under a different filtering regime. The current logging partially exposes this shift. In the current RC outside-option study, the mean predicted cost of the chosen bundle is 43.53 under full display, 43.42 under the strict-filter heuristic, and 43.36 under the no-filter heuristic. We also record displayed-offer ETA/IVT error: displayed ETA MAE is 2577 s under full display, 2564 s under the strict-filter heuristic, and 2384 s under the no-filter heuristic; the corresponding IVT MAE values are 2308 s, 2284 s, and 2131 s. These diagnostics are descriptive rather than causal, but they confirm that removing the filter changes the evaluated bundle distribution and should be interpreted jointly with the policy outcomes.

The empirical evaluation covers three instance types: the RC synthetic benchmark (Homberger–Gehring family) and the Austin and Seattle real-world-derived instances adapted from the Amazon Last Mile Routing Research Challenge [10], each with two bundled splits. While this provides cross-topology evidence spanning synthetic and real-world-derived geographies, all instances

share the same fleet size (15 vehicles, capacity 25) and the same demand intensity regime. The city studies are also low-sample: only two split pairs are available per city, so the Austin and Seattle results should be interpreted as suggestive external checks rather than definitive replication. Broader coverage — including larger fleets, additional real-world-derived instances, and comparison with published DRT assignment methods — is left as explicit future work.

The MNL utility parameters $(\beta, \bar{u}, u^{\text{home}}, \alpha_\tau, \alpha_\Delta, \sigma)$ are fixed design parameters of the simulator rather than values estimated from passenger survey data. This matters directly for interpretation. Under the original RC outside-option evaluation, the benchmark exhibits extremely low realized uptake of non-home bundles; under the calibrated higher-uptake regime, the same simulator structure yields materially higher acceptance. The paper therefore treats calibration as an evidence-setting decision rather than as a resolved demand-estimation result. The higher-uptake regime is useful because it restores behavioral variation, but it should not be read as a uniquely validated empirical demand model. Appendix Table A.13 records the current sensitivity evidence and the reproducible suite for filling the remaining low-, medium-, and higher-uptake cells under the current pipeline.

The seven scoring and menu parameters $(\mu_m, \mu_w, \mu_t, \mu_v, \mu_r, \mu_q, k)$ are similarly fixed design choices. Exact enumeration is exact only for the implemented surrogate objective on bounded candidate sets; it is not an exact solution to the full dynamic DRT problem with downstream HGS re-optimization. The pricing-baseline comparison is informative about one calibrated stress-test regime, but it is still a local result: flat markdown can outperform the Lambert-W-inspired rule in the reported higher-uptake environment, yet we do not claim that this ordering is invariant to other revenue targets, price caps, demand scales, or preference settings. A single-split sensitivity table from an earlier pipeline run is retained in Appendix Appendix C as an archival diagnostic, but it was not rerun under the current evaluation setup and should not be interpreted as confirmatory evidence that one solver, filter, or pricing rule robustly dominates the alternatives.

9. Conclusion

We formulated many-to-one DRT service-menu design as an assortment optimization problem over meeting-point and pickup-window bundles. The implemented framework combines MNL passenger choice, predicted route-aware bundle costs, common-offset pricing, exact enumeration for small candidate sets, and greedy forward selection for larger candidate sets. It also logs exact-vs-greedy approximation diagnostics so that the scalable solver can be evaluated against a surrogate benchmark.

The empirical contribution is a disciplined evidence program rather than a universal dominance claim. Exact-vs-greedy diagnostics quantify the cost of scalable menu construction; robust ETA-filtering comparisons show how eligibility rules reshape the feasible assortment before optimization acts; and low-, medium-, and high-uptake regimes clarify when profit, acceptance, non-home uptake, and welfare comparisons are behaviorally meaningful. Austin and Seattle remain external directional checks under a low-sample constraint.

The main practical lesson is that service-menu optimization should be managed as a layered system. Operators should audit ETA filtering before tuning downstream ranking, benchmark greedy approximations against exact enumeration where possible, and interpret pricing or welfare results only in regimes where passengers actually face live tradeoffs. Broader validation with additional city splits, empirically estimated demand parameters, and richer real-world-derived instances remains the next step.

References

- [1] Agatz, N., Erera, A., Savelsbergh, M., Wang, X., 2012. Optimization for dynamic ride-sharing: A review. *European Journal of Operational Research* 223, 295–303.
- [2] Ben-Akiva, M., Lerman, S.R., 1985. *Discrete Choice Analysis: Theory and Application to Travel Demand*. MIT Press.

- [3] Cordeau, J.F., Laporte, G., 2007. The dial-a-ride problem: Models and algorithms. *Annals of Operations Research* 153, 29–46.
- [4] Cortenbach, F., Kliewer, N., Masing, B., 2024. The dial-a-ride problem with meeting points: A problem formulation for shared demand-responsive transit. *Transportation Research Part C: Emerging Technologies* 166, 104757.
- [5] Czioška, P., Mattfeld, D.C., Sester, M., 2019. Real-world meeting points for shared demand-responsive transportation systems. *Public Transport* 11, 341–377.
- [6] Dong, J., Ye, H., Liang, H., 2020. Dial-a-ride problem with users’ accept/reject decisions based on service utilities. *Transportation Research Record* 2674, 308–323.
- [7] Fielbaum, A., Bai, X., Alonso-Mora, J., 2021. On-demand ridesharing with optimized pick-up and drop-off walking locations. *Transportation Research Part C: Emerging Technologies* 126, 103061.
- [8] Ho, S.C., Szeto, W.Y., Kuo, Y.H., Leung, J.M.Y., Petering, M., Tou, T.W.H., 2018. A survey of dial-a-ride problems: Literature review and recent developments. *Transportation Research Part B: Methodological* 111, 395–421.
- [9] McFadden, D., 1974. Conditional logit analysis of qualitative choice behavior. *Frontiers in Econometrics* , 105–142.
- [10] Merchan, D., Arora, J., Pachon, J., Konduri, K., Winkenbach, M., Parks, S., Noszek, J., 2022. 2021 Amazon last mile routing research challenge: Data set. *Transportation Science* 56, 1392–1407.
- [11] Parragh, S.N., Doerner, K.F., Hartl, R.F., 2008. A survey on pickup and delivery problems. *Journal für Betriebswirtschaft* 58, 21–51.
- [12] Rusmevichientong, P., Shen, Z.J.M., Shmoys, D.B., 2010. Dynamic assortment optimization with a multinomial logit choice model and capacity constraint. *Operations Research* 58, 1666–1680.

- [13] Stiglic, M., Agatz, N., Savelsbergh, M., Gradisar, M., 2015. The benefits of meeting points in ride-sharing systems. *Transportation Research Part B: Methodological* 82, 36–53.
- [14] Stumpe, M., Dieter, M., Friedrich, M., 2024. Designing taxi ridesharing systems with shared pick-up and drop-off locations: Insights from a computational study. *Transportation Research Part A: Policy and Practice* 185, 104102.
- [15] Talluri, K., van Ryzin, G., 2004. Revenue management under a general discrete choice model of consumer behavior. *Management Science* 50, 15–33.
- [16] Vidal, T., 2022. Hybrid genetic search for the cvrp: Open-source implementation and swap* neighborhood. *Computers & Operations Research* 140, 105643.
- [17] Wang, R., 2012. Capacitated assortment and price optimization under the multinomial logit model. *Operations Research Letters* 40, 492–497.

Appendix A. Evidence Hierarchy

Appendix B. Lambert-W Common-Offset Pricing Reference

We summarize the revenue-only common-offset reference problem used to motivate the implemented Lambert-W-inspired heuristic. This appendix is not a proof of optimality for the full simulator objective, which includes heterogeneous route costs, clipped prices, route-aware menu selection, and an outside option.

Setup.. Let $\mathcal{M}$ be the displayed menu. The operator sets a common price offset δ applied to all bundles, so that the price of bundle b is $p_b = \hat{c}(b) - R - \delta$, where R is the revenue target and $\hat{c}(b)$ is the predicted cost. The utility of bundle b at offset δ is

$$u_b(\delta) = u_b^0 + \beta (\hat{c}(b) - R - \delta),$$

where u_b^0 is the utility evaluated at zero price and $\beta < 0$ is the price sensitivity. With the outside option normalized to $u_0 = 0$, the MNL choice probability is

$$\pi_b(\delta) = \frac{e^{u_b(\delta)}}{e^{u_0} + \sum_{b' \in \mathcal{M}} e^{u_{b'}(\delta)}},$$

Reference objective.. Following the revenue-management setting of Talluri and van Ryzin [15], consider the expected displayed revenue under the shared offset,

$$\mathcal{R}(\delta) = \sum_{b \in \mathcal{M}} \pi_b(\delta) p_b(\delta).$$

The outside option remains in the denominator through e^{u_0} , so the total probability assigned to displayed bundles is generally below one. The full simulator objective used in the experiments is instead an expected net-profit surrogate,

$$\hat{\Pi}(\mathcal{M}) = \sum_{b \in \mathcal{M}} \pi_b(\mathcal{M}) \{p_b - \hat{c}_{\text{sys}}(b)\},$$

as defined in Eq. 6. We do not simplify that objective to a linear expression in δ ; doing so would incorrectly remove the outside-option denominator and would also ignore the route-cost heterogeneity that the simulator evaluates.

Lambert-W closed form.. When the objective is the expected revenue $\sum_b \pi_b(\delta) p_b$ (equivalently, when costs are treated as fixed), the first-order condition with respect to δ yields a transcendental equation solvable via the Lambert W function. This revenue-based derivation motivates the implemented common-offset heuristic, but it should not be read as a proof that the clipped pricing rule is globally profit-optimal under the full evaluated objective. Define

$$S = e^{u_0} + \sum_{b \in \mathcal{M}} e^{u_b^0 + \beta(\hat{c}(b) - R)}.$$

Under the shared-offset parameterization, the first-order condition can be rearranged into the standard Lambert-W form

$$we^w = S/e,$$

with solution $w = W_0(S/e)$ on the principal branch. Mapping back to the price offset gives

$$\delta^* = \frac{W_0(S/e) + 1}{\beta}. \quad (\text{A.1})$$

The price for each bundle is then clipped to the platform bounds:

$$p_b = \text{clip}(\hat{c}(b) - R - \delta^*, p_{\min}, p_{\max}).$$

This appendix records the closed-form motivation for a narrow reference problem only. The implemented policy in the main text adds route-aware menu selection, heterogeneous predicted costs, price clipping, and downstream HGS re-optimization, so Eq. (A.1) should be interpreted as a heuristic reference transform rather than an exact optimality certificate for the final simulator objective.

Conditions for validity. The common-offset assumption requires that all passengers share the same price sensitivity β and that utility is additively separable in price. Both hold by construction in the MNL simulator (Table 2). The Lambert-W solution is real-valued whenever $S > 0$, which is guaranteed as long as at least one bundle has finite utility. The home bundle’s large positive utility term ($u^{\text{home}} = 3.2$) ensures $S \gg 0$ in all practical cases, which is why removing the home bundle from the pricing set causes S to collapse and δ^* to fall below $p_{\min}$, collapsing all OOH prices to the floor.

Appendix C. Scoring-Weight Sensitivity Analysis

Table C.15 reports an *archival* single-split diagnostic from an earlier pipeline configuration: the net-profit gap of the no-filter heuristic versus full display when each of the seven scoring and menu parameters is individually perturbed by $\pm 50\%$ from its baseline value on the `rc_0_to_1` split. These numbers are retained only to show relative directional sensitivity within that historical setting; they are not directly comparable to the current outside-option results in Section 6.

Within that historical setting, the route-delay penalty weight μ_r and the capacity-risk penalty weight μ_q exhibit the largest downward sensitivity when reduced to 50% of their baseline values. The IVT weight μ_v is insensitive across the tested range, reflecting the low variance in predicted in-vehicle times for the RC instance. The menu size k shows a modest negative effect at $k = 5$, likely

because larger menus include marginal bundles that dilute the choice probability of the highest-profit options. Re-running this sweep under the current evaluation pipeline remains future work.

Appendix D. Predictor Validation

Table D.16 reports the mean absolute error (MAE) of the CNN-2d predictor and a naive mean-prediction baseline for each output head, evaluated on a held-out 20% partition of the training buffer across all six RC splits.

The naive mean baseline predicts the training-set mean for each head; it serves as the simplest possible zero-information predictor. The CNN-2d reduces cost prediction error by 15.3% relative to this baseline. The cost head is the primary input to the system-evaluation and menu-value calculations in Eqs. 5 and 6, so improvement in cost prediction directly benefits menu ranking quality.

For ETA and IVT, the CNN MAE is higher than the naive baseline because the predictor is trained online on data collected during early training episodes, when the routing state is highly variable. In deployment, ETA and IVT predictions are blended with heuristic estimates derived from cheapest-insertion cost computations (blend weight 0.35 CNN, 0.65 heuristic; parameter `menu_time_prediction_blend`), which substantially reduces the effective prediction error applied to the score function.

Appendix E. Split-Level Uncertainty and Policy-Conditioned Diagnostics

Table 10 reports the Phase 19 uncertainty summary used in the main text. The sampling unit throughout is the split-level paired mean net-profit gap relative to full display. RC therefore supports modest split-level uncertainty reporting with $n = 6$ split pairs, whereas Austin and Seattle remain descriptive because each city contributes only $n = 2$ split pairs.

Table E.17 lists the six RC split-level paired gaps together with their within-split episode-level CI half-widths. These within-split intervals are useful for

diagnosing heterogeneity across the replayed traces inside a fixed split, but the paper’s uncertainty statements are based on the split-level paired means rather than on pooling episodes across splits.

Table E.18 gives the corresponding Austin and Seattle split-level paired gaps. With only two split pairs per city, these numbers should be read as descriptive evidence about directional stability rather than as precise interval estimates.

Finally, Table E.19 separates policy-conditioned displayed-offer diagnostics from realized passenger-facing outcomes. The displayed ETA/IVT MAE and false-negative pruning columns diagnose how each policy changes the offered-candidate distribution seen by the predictor. Acceptance rate and consumer surplus, by contrast, remain realized evaluation outcomes under replayed traces.

Appendix F. Computational Cost

Table F.20 summarises the wall-clock computational cost of the three main phases: predictor training, menu-construction inference, and HGS back-end routing.

The menu-construction inference time (44.9 ms per request) includes the CNN forward pass and the cheapest-insertion cost computation used to estimate marginal routing cost. HGS routing is invoked periodically (every `reopt` steps) rather than at every request; the mean step time of ≈ 43 ms reflects the amortised cost across steps with and without routing calls.

Appendix G. Reproducibility

The project keeps the experiment logic and the paper layer synchronized through a small set of public commands:

```
python scripts/run_study.py --study rc_main_optout
python scripts/run_study.py --study phase22_eta_compare
python scripts/run_study.py --study phase22_uptake_calibration
python scripts/run_study.py --study phase23_pricing_baselines
```

```
python scripts/run_study.py --study phase29_exact_greedy_gap
python scripts/run_study.py --study phase30_robust_filtering
python scripts/run_study.py --study phase31_uptake_menu_value
python scripts/run_study.py --study austin_main
python scripts/run_study.py --study seattle_main
python scripts/extract_phase22_results.py
python scripts/extract_phase23_results.py
python scripts/extract_phase24_results.py
python scripts/build_artifacts.py --study rc_main_optout
python scripts/build_artifacts.py --study phase29_exact_greedy_gap
python scripts/build_artifacts.py --study phase30_robust_filtering
python scripts/build_artifacts.py --study phase31_uptake_menu_value
python scripts/build_artifacts.py --study austin_main
python scripts/build_artifacts.py --study seattle_main
python scripts/build_manuscript.py
```

For local smoke verification, the repository also ships a minimal manifest:

```
python scripts/run_study.py --study smoke_rc
```

The manuscript consumes generated tables and figures from `artifacts/`, while full raw experiment outputs remain under `outputs/` and are excluded from version control.

Table 1: Main notation. Symbols are listed in order of introduction; units are given where applicable.

Symbol	Definition
r	Passenger request
$\mathcal{B}(r)$	Feasible bundle set for request r
b	= Service bundle: meeting point, time window, price
$(m_b, [t_b^{\text{lo}}, t_b^{\text{hi}}], p_b)$	
b^0	Home pickup bundle (zero walking distance)
$\mathcal{M}(r)$	Displayed menu, $\mathcal{M}(r) \subseteq \mathcal{B}(r)$, $ \mathcal{M}(r) \setminus \{b^0\} \leq k$
k	Cap on displayed non-home offers
$\hat{c}(b)$	Predicted marginal cost of bundle b
$c(b)$	Realized marginal operational cost of serving bundle b
$\hat{c}_{\text{sys}}(b)$	Route-aware surrogate cost used in menu scoring
d_b	Walking distance to meeting point m_b (metres)
$\hat{\tau}_b$	Predicted pickup ETA for bundle b
$\hat{\tau}_b^{\text{acc}}$	Passenger’s acceptable ETA upper bound
$\hat{t}_b$	Predicted in-vehicle time for bundle b
$\Delta_b = \hat{\tau}_b - t_r^{\text{req}}$	Pickup time deviation from passenger’s requested time
σ	Distance scale parameter in walking-disutility term (metres)
β	Passenger price sensitivity ($\beta < 0$)
$\mu_m, \mu_w, \mu_t, \mu_v$	Non-negative scoring weights (margin, walk, time, IVT)
u_b	Passenger utility for bundle b (MNL)
u_0	Outside-option utility, normalized to 0 in the MNL model
$\pi(b \mid \mathcal{M})$	MNL choice probability for b given menu $\mathcal{M}$
Π	Aggregate net profit over all requests
T	Planning horizon (set of decision epochs)
$t_b^{\text{lo}}, t_b^{\text{hi}}$	Lower and upper bounds of the pickup time window for bundle b
p_b	Price charged to passenger for bundle b (monetary units)
δ_{impl}	Implemented common-offset pricing parameter (Eq. 3)
R	Operator revenue target (monetary units)
$p_{\min}, p_{\max}$	Platform price floor and cap (monetary units)
$\bar{u}$	Base utility intercept (MNL)
u^{home}	Additional utility of the home bundle (MNL)
α_τ	In-vehicle time weight (MNL)
α_Δ	Pickup-time deviation weight (MNL)

Table 2: MNL utility parameter values used in the experiments.

Symbol	Description	Value	Source
β	Price sensitivity	−0.25	Simulator design parameter
$\bar{u}$	Base utility intercept	−2.0	Simulator design parameter
u^{home}	Home-bundle utility bonus	3.2	Simulator design parameter
α_τ	In-vehicle-time weight	−0.002	Simulator design parameter
α_Δ	Pickup-time-deviation weight	−0.0015	Simulator design parameter
σ	Walking-distance scale	1.0 (instance-distance units)	Simulator design parameter
R	Revenue target	50.0	Simulator design parameter
$p_{\min}$	Minimum price (floor)	−6.0	Platform constraint
$p_{\max}$	Maximum price (cap)	5.0	Platform constraint

Table 3: Exact-vs-greedy assortment diagnostics on the RC benchmark.

Solver	Gap vs full	Gap vs exact	Rel. opt. gap	Menu overlap	Exact time	Greedy time	Logged share
full display	—	—	0.0000	0.000	0.0000	0.0000	0.000
exact bench- mark	0.000	—	0.0000	1.000	0.0372	0.0032	1.000
greedy approx- imation	0.000	0.000	0.0000	1.000	0.0392	0.0033	1.000
auto solver	0.000	0.000	0.0000	1.000	0.0379	0.0032	1.000

Table 4: Robust ETA-filtering comparison on the RC benchmark.

Filter	Gap vs full	FN pruning	Home-only share	Menu size	Acceptance	Non-home acc.
full display	—	0.992	0.918	1.085	0.000	0.000
hard ETA filter	0.000	0.992	0.918	1.082	0.000	0.000
calibrated ETA filter	0.000	0.992	0.918	1.082	0.000	0.000
interval-overlap filter	0.000	0.992	0.918	1.082	0.000	0.000
no ETA filter	0.000	0.000	0.000	2.094	0.000	0.000

Table 5: Menu-value comparison across low, medium, and high uptake regimes.

Regime	Policy	Gap vs full	Acceptance	Non-home acc.	Surplus	FN pruning	Floor hit
low	full display	–	0.000	0.000	0.000	0.992	0.000
low	exact solver	0.000	0.000	0.000	0.000	0.992	0.000
low	auto solver	0.000	0.000	0.000	0.000	0.992	0.000
low	no ETA filter	0.000	0.000	0.000	0.000	0.000	0.000
low	flat markdown	0.000	0.000	0.000	0.000	0.992	0.000
medium	full display	–	0.002	0.002	-0.250	0.991	0.167
medium	exact solver	0.169	0.001	0.001	-0.125	0.992	0.083
medium	auto solver	0.169	0.001	0.001	-0.125	0.992	0.083
medium	no ETA filter	2.001	0.001	0.001	-0.125	0.000	0.083
medium	flat markdown	0.500	0.002	0.002	-0.125	0.991	0.000
high	full display	–	0.009	0.009	-0.500	0.991	0.333
high	exact solver	0.112	0.009	0.009	-0.625	0.992	0.417
high	auto solver	0.112	0.009	0.009	-0.625	0.992	0.417
high	no ETA filter	-0.030	0.007	0.007	-0.250	0.000	0.167
high	flat markdown	1.575	0.007	0.007	-0.188	0.991	0.000

Table 6: RC Main Opt-out Validation policy comparison.

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-4377.492	–	–	2.151	1.151
strict-filter	-4373.527	3.965	0.389	1.573	0.573
heuristic					
no-filter heuristic	-4370.880	6.612	0.444	2.361	1.361

Table 7: Austin (Amazon Last Mile) Main Policy Comparison

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-2992.645	–	–	9.037	8.037
strict-filter	-2998.841	-6.196	0.333	2.458	1.458
heuristic					
no-filter heuristic	-2960.523	32.122	0.833	2.764	1.764

Table 8: Seattle (Amazon Last Mile) Main Policy Comparison

Policy	Mean net profit	Gap vs full	Win rate vs full	Mean menu size	Avg meeting-point count
full display	-3149.981	–	–	5.354	4.354
strict-filter	-3103.059	46.921	1.000	1.658	0.658
heuristic					
no-filter heuristic	-3098.024	51.957	0.917	2.315	1.315

Table 9: City-level management summary for the no-filter heuristic against full display and the strict-filter heuristic.

City	No-filter vs full gap	No-filter vs strict gap	Acceptance delta	Consumer-surplus delta
Austin	32.122	38.318	-0.118	0.224
Seattle	51.957	5.035	-0.047	0.132
pooled	42.039	21.677	-0.083	0.178

Table 10: Split-level paired uncertainty summary for the strict-filter and no-filter heuristics against full display. The inferential unit is the split-level paired mean net-profit gap; RC uses six split pairs, while Austin and Seattle each use two split pairs and are therefore descriptive only.

Study	Policy	Mean gap	Split-level interval/range	Interval type	Positive-split share	Gap as % of [full]	Sampling unit
RC	outside- strict-filter heuristic	3.97	[-6.46, 12.07]	split bootstrap interval	0.83	0.09	6 split pairs
option							
RC	outside- no-filter heuristic	6.61	[1.41, 11.01]	split bootstrap interval	0.83	0.15	6 split pairs
option							
Austin	strict-filter heuristic	-6.20	[-26.25, 13.86]	observed range	0.50	-0.21	2 split pairs (descriptive)
Austin	no-filter heuristic	32.12	[24.05, 40.20]	observed range	1.00	1.07	2 split pairs (descriptive)
Seattle	strict-filter heuristic	46.92	[43.06, 50.78]	observed range	1.00	1.49	2 split pairs (descriptive)
Seattle	no-filter heuristic	51.96	[50.26, 53.65]	observed range	1.00	1.65	2 split pairs (descriptive)

Table 11: Benchmark-role bridge summary separating the degenerate RC mechanism benchmark from the city impact benchmarks. Gate status is evaluated on the full-display reference and the no-filter heuristic; reference-pass-only means the benchmark itself is behaviorally non-degenerate even if the no-filter policy falls outside the gate.

Study	Role	Gate status	Full acceptance	No-filter acceptance	Full opt-out	No-filter gap	Gap as % of [full]	Bridge use
RC	outside- mechanism	degenerate	0.001	0.000	0.994	6.61	0.15	mechanism only
option								
Austin	impact reference	pass only	0.161	0.044	0.694	32.12	1.07	primary impact
Seattle	impact reference	pass only	0.060	0.013	0.850	51.96	1.65	supporting check

Table A.12: Evidence hierarchy used to interpret the empirical results. The table separates mechanism diagnostics, behavioral stress tests, descriptive external checks, and archival diagnostics so that no claim is stronger than its evidence tier.

Evidence tier	Source	Supports	Does not support
Mechanism diagnostic	RC outside-option benchmark and ETA-identification study	Candidate-set distortion, filter behavior, displayed-offer diagnostics	Strong operator- or demand-calibration claims
Behavioral stress test	Higher-uptake RC calibration and pricing/welfare studies	Pricing and tradeoffs when acceptance is behaviorally live	External validation of the true passenger demand model
Descriptive external check	Austin and Seattle two-pair city studies	Directional consistency real-world-derived geographies	con- across mates or broad generalization
Archival diagnostic	Earlier sensitivity and development sweeps	Implementation history and directional checks	Confirmatory robustness under the current pipeline

Table A.13: MNL-regime sensitivity summary. Rows marked current-evidence baseline are extracted from completed v1.6 outputs; completed-current-pipeline rows are produced by the Phase 27 sensitivity manifests. Higher-uptake rows should be read as stress-test evidence rather than externally validated demand calibration.

Regime	Policy/pricing rule	Evidence status	Gap vs full	Acceptance	Non-home acc.	Surplus	Floor hit
low	strict-filter heuristic	current evidence	0.81	0.004	0.001	-1.083	–
higher	menu optimization higher-uptake regime	current evidence	4.98	0.165	0.038	-1.338	–
higher	lambertw pricing	current evidence	-3.07	0.165	0.037	-1.340	0.922
higher	cost-plus pricing	current evidence	-26.70	0.168	0.039	-1.448	0.980
higher	flat markdown pricing	current evidence	163.50	0.145	0.037	-0.750	0.000

Table A.14: Blended-filter validity diagnostics on the RC mechanism benchmark. Displayed-offer errors are computed over all displayed non-home offers, while selected-offer errors are weakly identified in the low-uptake RC benchmark because few non-home offers are accepted.

ETA/filter signal	FN pruning	Displayed ETA MAE	Selected ETA MAE	Displayed IVT MAE	Selected IVT MAE	Acceptance	Gap vs full
deployed blended ETA	0.884	2582.1	147.1	2289.9	117.8	0.004	0.81
heuristic ETA	0.864	0.0	0.0	1976.8	186.5	0.005	-1.87
stronger calibrated ETA	0.864	466.2	61.2	2058.6	248.9	0.006	-6.22
oracle ETA filter proxy	0.864	2237.8	107.2	1948.0	69.5	0.004	2.05

Table C.15: Archival single-split sensitivity diagnostic from an earlier pipeline configuration. Baseline gap in that setting is +659.6; the table is included only to show relative directional sensitivity within that historical setup, not to support the main claims of the paper.

Parameter	Description	-50%	Baseline	+50%
μ_m	Margin weight	659.6	657.7	663.2
μ_w	Walking-distance weight	663.2	657.9	661.5
μ_t	Pickup-time deviation weight	656.0	657.7	659.4
μ_v	In-vehicle time weight	659.6	659.6	659.6
μ_r	Route-delay penalty weight	578.0	659.6	657.5
μ_q	Capacity-risk penalty weight	597.5	659.6	645.6
k	Menu size cap ($k = 2, 3, 5$)	607.8	659.6	637.8

Table D.16: Predictor MAE on held-out buffer data (mean $\pm$ std across 6 RC splits). CNN-2d reduces cost MAE by 15.3% relative to the naive mean baseline. ETA and IVT CNN MAE is higher than the naive baseline because the CNN is trained on noisy online data; in deployment, ETA and IVT predictions are blended with heuristic estimates (blend weight 0.35 CNN, 0.65 heuristic).

Predictor	Cost MAE (monetary units)	ETA MAE (seconds)	IVT MAE (seconds)
CNN-2d	5.00 ± 0.59	$5,703 \pm 286$	$5,275 \pm 247$
Naive mean	5.91 ± 0.57	$1,863 \pm 66$	$1,860 \pm 67$

Table E.17: RC outside-option split-level paired gaps against full display. Within-split CI half-width is computed from episode-level pairing inside each split; main-text uncertainty uses the split-level paired means rather than pooled episode rows.

Split pair	Strict gap	Strict within-split CI/2	No-filter gap	No-filter within-split CI/2
0->1	0.35	19.60	5.08	10.97
0->1 (seed 2025)	6.78	21.15	7.63	24.29
1->0	-19.85	13.06	-5.31	10.82
2->3	16.06	20.43	13.19	19.96
3->2	13.32	17.64	6.12	14.08
4->2	7.13	30.09	12.96	23.57

Table E.18: Austin and Seattle split-level paired gaps against full display. Each city has only two split pairs, so these rows should be read descriptively rather than as stable interval estimates.

City	Split pair	Strict gap	No-filter gap
Austin	0->1	-26.25	24.05
Austin	1->0	13.86	40.20
Seattle	0->1	50.78	53.65
Seattle	1->0	43.06	50.26

Table E.19: City-level policy-conditioned displayed-offer diagnostics. Displayed ETA/IVT MAE and false-negative pruning are diagnostics on displayed offers rather than direct welfare outcomes; acceptance rate and consumer surplus are realized passenger-facing outcomes.

City	Policy	FN pruning	Displayed ETA MAE	Displayed IVT MAE	Acceptance rate	Consumer surplus
Austin	full display	0.598	598.5	726.8	0.161	-1.227
Austin	strict-filter heuristic	0.598	570.8	707.2	0.043	-1.102
Austin	no-filter heuristic	0.000	802.5	951.0	0.044	-1.003
Seattle	full display	0.782	943.3	727.2	0.060	-1.315
Seattle	strict-filter heuristic	0.782	908.4	689.8	0.010	-1.194
Seattle	no-filter heuristic	0.000	3847.2	2502.7	0.013	-1.183

Table F.20: Computational cost summary for the RC instance (90 customers, 10 meeting points, 15 vehicles). Training and inference times are measured on a standard CPU (no GPU used). HGS time limits are upper bounds configured per episode; actual routing time is typically below the limit.

Phase	Metric	Value
Predictor training	Training episodes	12
	Steps per episode	≈ 700
	Wall-clock per episode	≈ 26.5 s
	Total training time	≈ 318 s (≈ 5.3 min)
	Initial phase epochs	12
	Initial phase total time	≈ 0.4 s
Menu-construction inference	Mean time per request	44.9 ms
	(CNN forward pass + insertion cost)	
HGS routing	Reopt time limit (per step)	1.1 s
	Final reopt time limit	1.5 s
	Mean step time (menu + HGS)	≈ 43 ms